

Understanding economic growth in India, A prerequisite

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Abstract

We follow a recent development in the estimation and testing of multiple structural breaks in linear models to identify phases of growth in India since 1950. A noteworthy feature of the methodology is to ‘allow the data to parametrise the model’ yielding results that are immune to the prior beliefs of the researcher. Our estimates reveal that there have been two growth regimes in India since 1950. Next we undertake a decomposition by sector of the contribution to the change in the growth rate across these two regimes. Finally, by means of a simple econometric model, we test a hypothesis that emerges from our results to provide an explanation of this growth transition. It leads us to conclude that recognition of a services-led growth is a prerequisite to understanding accelerated economic growth in India. In passing, we consider the bearing of our results on other explanations of the same transition that have been provided.

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"If you torture the data enough, nature will always confess."

Ronald Coase

I. Introduction

In this paper we adopt a recent development in the econometrics of estimating and testing multiple structural breaks in linear models to throw light on a topic of some interest to researchers¹ currently, the engine of growth in India. The main issue of contention is the turning point(s) of economic growth since the nineteen fifties and what has contributed to it. Mostly, the authors have identified significant changes in economic policy and tested for whether they have succeeded in bringing about the growth shift, even though there is no agreement on the precise policies that did so. Our own approach is outright empiricist by comparison. We adopt a methodology that identifies growth regimes endogenously rather than one based on exogenous information such as the date of initiation of a policy change. This immunises² the methodology to a researcher's preconception regarding the timing of growth shifts and their likely causes. The methodology that we use is the one developed by Bai and Perron (1998) to estimate and test for multiple structural breaks in a time series. This yields us results that we believe enhances our understanding of economic growth in India for the close to half a century just passed.

II. Methodology and data

The growth rates of aggregate and sectoral GDP may be estimated using the exponential function $\ln Y_t = a + gt + u_t$, where $\ln Y$, g , t and u , respectively, denote the log of income, growth rate, time trend and random disturbance term. The subscript t indicates the time point. The parameters of the above regression function - a and g - would vary from one growth regime to another, making it necessary to identify the change point. Therefore, we first estimate the breakdates of the above model for aggregate and sectoral series and use them to estimate the period-wise growth rates. The methodology for estimating the breakdates is explained in detail below.

The exponential growth model containing $m+1$ growth regimes and m break dates ($T_1, \dots, T_m$) can be written as follows:

¹ See Fischer (2003), Nagaraj (2006) Baldev Raj Nayyar (2005), Deepak Nayyar (2006), Panagariya (2005), and Rodrik and Subramanian (2006).

² See Hansen (2001) for an accessible discussion of developments in the associated econometrics.

$$\begin{aligned}
\ln Y_t &= a_1 + g_1 t + u_t, & t &= 1, \dots, T_1 \\
\ln Y_t &= a_2 + g_2 t + u_t, & t &= T_1 + 1, \dots, T_2 \\
&\vdots \\
\ln Y_t &= a_{m+1} + g_{m+1} t + u_t, & t &= T_m + 1, \dots, T.
\end{aligned} \tag{1}$$

Here we adopt the convention that $T_0 = 0$ and $T_{m+1} = T$ the total number of observations. The number of break points m and the break dates $(T_1, \dots, T_m)$ are treated as unknown and estimated from the data. There are two methodologies to estimate the breakdates endogenously: (i) the instability-test based approach (Andrews 1993, Andrews and Ploberger 1994) and (ii) the regression-based least-squares approach (Bai 1997, Bai and Perron 1998).. A problem with the instability-test procedure is that the distribution of the F-statistic used to test the null of constancy of the parameters is derived for non-trending regressors. Hence, it is not applicable here as our model contains a trended regressor. Therefore, we use the Bai and Perron (1998, 2003) method. This is based on the least squares principle common to regression analysis and allows the simultaneous estimation of multiple breaks. It is an improvement over the instability test as the latter can be used to select only one breakdate at a time. The breakdates are estimated as global minimisers of the sum of squared residuals from an OLS regression of (1) using a dynamic programming algorithm (see Bai and Perron 2003). The procedure is as follows. For each m partition $(T_1, \dots, T_m)$ denoted $\{T_p\}$, the associated least squares estimates $\beta_p = (a, g)_p$ are obtained by minimising the sum of squared

residuals $\sum_{j=1}^{m+1} \sum_{t=T_{j-1}+1}^{T_j} [\ln Y_t - a_j - g_j t]^2$. Let $\hat{\beta}_p(T_p)$ denote the resulting estimates.

Substituting them into the above objective function and denoting the resulting sum of squared residuals as $S_T(T_1, \dots, T_m)$, the estimated break points $(\hat{T}_1, \dots, \hat{T}_m)$ are such that $(\hat{T}_1, \dots, \hat{T}_m) = \underset{(T_1, \dots, T_m)}{\operatorname{argmin}} S_T(T_1, \dots, T_m)$, where the minimisation is over all possible partitions $(T_1, \dots, T_m)$ such that $T_i - T_{i-1} \geq h$. Note that h is the minimum length assigned to a segment and T_i is the i^{th} break point. The procedure considers all possible combination of segments and selects that partition which minimises the sum of squared residuals. Thus the least-squares estimates of breakdates are those that minimise the full-sample sum of squared residuals in (1). Note that estimated breakdate is consistent even in the presence of non-stationarity of the regressors and disturbances³. Finally, Model (1), which we have used, has

³ See Bai (1997).

the advantage that the theoretical properties of the estimated break points in such a model have been examined most extensively⁴.

The above procedure estimates sequentially the optimal partition of the sample starting from two to the maximum allowed by T and h . When the number of break points is unknown, a test based on the *supF* statistic has been proposed (Bai and Perron 1998), to choose the number of breakpoints. Once again, this testing procedure assumes non-trending regressors and is therefore inapplicable in the present context. An alternative approach which accommodates trending regressors is to use the Bayesian Information Criteria (BIC). This has been demonstrated to be superior to other information criteria in the determination of the number of structural breaks (Wang, 2006). Here the number of breaks selected is that for which BIC is at a minimum. We adopt this procedure to choose the number of breaks⁵. The criterion is particularly appropriate when multiple breaks are considered because it introduces a penalty factor for additional break points which necessarily reduces the sum of squared residuals⁶, as is apparent from below:

$$BIC(m) = \ln \hat{\sigma}^2(m) + p^* \ln(T)/T$$

$$p^* = (m + 1)q + m + p$$

$$\hat{\sigma}^2 = T^{-1} \sum_{t=1}^T \hat{u}_t^2$$

where m is the number of breaks, q is the number of explanatory variables whose coefficients are subjected to shift and p is the number of explanatory variables whose coefficients are constant.

Before proceeding to the estimation we consider another issue. This relates to the question of whether we are faced with what has been referred to in the literature as a ‘partial’ versus ‘pure’ structural break. In the present context this translates into the following question. We are concerned with a change in the rate of growth of output which implies a shift in the slope coefficient. When this is accompanied by an intercept shift in the data generating process, it is important to allow for this level shift in the estimation of the breakdate. Simulation results of Perron and Zhu (2005) show that incorporation of a level shift of more than 0.3 in absolute value improves the accuracy of breakdate estimation. However, a practical problem is that the estimated change in the intercept cannot be a good

⁴ See Perron and Zhu (2005).

⁵ For a similar application see Uctum et al (2006).

⁶ See Bai and Perron (2003) for a comparison of the performance of BIC and *supF* statistic.

guide due to contamination and feedback effects between the estimated level shifts and break dates. Here, along with an examination of the estimated intercept shift

a visual comparison of the actual and fitted series has been suggested to verify whether the intercept shift is a true level shift rather than a random deviation away from the trend line (Perron and Zhu 2005). We adopt this approach.

The study covers the period from 1950-51 to 2003-04. The choice of time period follows from two considerations. First, we are interested in growth regimes after the launching of a development programme in independent India so that we may track the effect of changes in the policy regime. Secondly, it has already been established⁷ that when the entire twentieth century is taken the dominant break occurs around the year 1950. Data on Gross Domestic Product (GDP) at aggregate and sectoral level and sectoral shares in total GDP up to 1999-2000 were obtained from *National Account Statistics of India*, an electronic database supplied by the Economic and Political Weekly Research Foundation and the data for the remaining years were taken from *National Account Statistics*, published by the Central Statistical Organisation (CSO). The level of disaggregation adopted in this study is determined by the data availability. All the figures are in 1993-94 prices.

In our estimation of the breakdates the minimum length of a segment, h , has been fixed at 8. This implies a maximum of five breaks or six growth regimes in our sample extending over 1950-51 to 2003-4. A trimming of 15 per cent of the total observations, implied by $h = 8$, is considered appropriate for a sample size of the present study (see Bai and Perron 2003). Also, a minimum of 8 years per segment seems reasonable to us when studying long-run growth. We are, however, aware that an element of judgment is involved here⁸. Now the search for possible break would be confined to the period 1957-58 to 1996-97. The growth rates across regimes of a series are estimated by imposing kinks at the estimated breakpoints using the procedure due to Boyce (1986). This procedure maintains the continuity of the exponential trend line at the break points.

III. Results and their interpretation

The estimated breakdates⁹ are presented in Table 1 and the growth rates for the sub-periods in Table 2. The breakdate has been estimated allowing for a shift in the intercept alongside a

⁷ See Hatekar and Dongre (2005).

⁸ Our sensitivity analysis shows that estimated breakdates are invariant to h of four and five, given the number of breaks estimated using $h = 8$.

⁹ Estimation of the breakdates were done using the software package *strucchange* in R written by Zeileis A, Leisch F, Hansen B, Hornik K, Kleiber C and Peters. For details see Zeileis et al. (2005).

change in the slope coefficient, the case of a pure structural change. Our visual comparison of the fitted series and actual series showed that in almost all cases the change in the slope is accompanied by a shift in the intercept¹⁰. Furthermore, in the overwhelming majority of instances (50 intercept shifts out of a total 71 for the 24 series) the change in the intercept exceeded the absolute value 0.3 found by Perron and Zhu (2005) to be the critical value beyond which the accuracy of the breakdate estimates can be affected if the intercept shift is not allowed for¹¹.

[Table 1 and 2 go about here]

1. Our first finding is that the growth rate of GDP shows only one shift, in 1978-79. This is broadly in line with the findings of other researchers who have employed, as we have done, the methodology of determining structural break without an arbitrary partitioning of the time series based on priors held. It is noticeably close to the result reported by Wallack (2003) that the GDP series breaks in 1978-79.
2. We find that all the main components of GDP show acceleration at some point during the period of analysis. This involves not only the categories of Primary, Secondary and Tertiary but also their components up to the next level of disaggregation available. This finding is at odds with the reported finding of Wallack that no trend break can be established in the disaggregated GDP series. If true, that would imply that there is no underlying dynamism to growth in the economy. That is, all the sectors grow at a constant rate. However, the growth rate of aggregate GDP yet accelerates as the fast(er) growing sectors come to occupy larger shares of the economy. An acceleration of the overall growth rate when the sectoral growth rates remain constant is purely an artifact of accounting, due to the reallocation of output from the slow-growing to the fast-growing sectors of the economy. Our estimates conclusively establish that this is not so. We find change in the trend growth rate across all sectors in the economy. Not only is this plausible, it is also in keeping with conventional wisdom among researchers of the Indian economy that shifts in the Indian growth data such as ‘the Green Revolution’ and ‘Industrial stagnation since the mid-sixties’ represent trend breaks.

¹⁰ To save space we present a representative graph of fitted and actual series of GDP in the appendix Figure A1. The figures for the remaining series are available from us upon request.

¹¹ We are aware of an additional empirical issue. In the case of multiple breaks it is conceivable that there is an intercept shift between some segments and not between others. However, our methodology does not allow us to accommodate this forcing us to allow for an intercept shift for every break in a series.

3. Agricultural growth shows a trend break in 1964-65 when growth accelerates. As the estimates of a structural break come with an associated confidence interval we need not hold fast to the point estimate and we are open to the interpretation that the break takes place “sometime in the mid-sixties”. However the point estimate does suggest the interpretation that the acceleration of agricultural growth is not entirely due to the miracle seeds with which the Green Revolution tends to be identified. It may also owe something to the steady expansion in irrigated area in the decade and half preceding the mid-sixties. Clearly, the methodology of generating structural breaks in the data endogenously has proved decisive in this understanding. The traditional practice is to measure the rate of growth of agriculture prior to 1965-66 and after 1966-67, these two years - being ones of severe drought when output was well below trend - having been excluded from the time series on output in both segments. The usual result is to find the rate of growth higher in the second period and to conclude that the acceleration has occurred in 1967-68. This is interpreted as the impact of the Green Revolution, the principal feature of which is seen as the spreading use of high-yielding varieties of seed and the application of fertiliser both introduced in the years of food shortage in the mid 1960s. It is, however, difficult to assume that this package alone could have raised the rate of growth permanently. Our finding here that the growth rate of the agricultural sector accelerates from 1964-65 suggests that the conventional finding of an acceleration of agricultural growth from 1967-68 is an artifact of the method of using exogenous information to partition the time series. We consider it an intuitive appeal of our estimate that the growth acceleration is taking place in a year of fast growth. The data show that in 1964-65 agricultural output increases by over 10 percent.

4. The first structural break in the series for Manufacturing output occurs in the mid-sixties, when growth shifts downwards. Again, it is intuitive that the break occurs in a year in which agricultural output, presumably a major determinant of manufacturing output growth, declines very substantially. Two further structural breaks follow. The first of these is positive and occurs in 1982-83 while the second is negative and occurs in 1994-95. Both these dates have an economic significance. Let us consider the second first as the interpretation of this fact is straightforward. Here, the conclusion can only be that the substantial liberalisation of the economy in the years following 1991 has not succeeded in raising the rate of growth of manufacturing at least till 1996-97 for which a break can be allowed for in keeping with the choice of $h = 8$. The break of 1982-83 is of greater significance in our search for an explanation of faster growth in India. Despite an acceleration, it comes a full four years after

the recorded break in growth of GDP in 1978-79. Even allowing for a confidence interval, note that the take off in the aggregate growth rate is definitely prior to (see Table 2) that of manufacturing. So, altogether, we find *prima facie* evidence against the possibility that manufacturing has led the acceleration in the growth rate. To us this is entirely plausible from a statistical point of view. The contribution to the overall growth rate by a sector is the product of the growth rate of the sector into its share of output. Note from Table 1 that it is only registered manufacturing that accelerates in 1982-83, unregistered manufacturing doing so only in the mid-eighties. The share of registered manufacturing in the GDP in 1982-83 was a mere 8.7 percent. Therefore it is unlikely to have lifted the aggregate growth rate given the extent of the acceleration involved (see Table 2).

5. Following from the comment above on our estimates of breaks in the growth of manufacturing output we conclude that the explanation often encountered that it is trade and industrial policy reforms that have resulted in a faster rate of growth of the economy is off the mark. First, when the rate of growth of manufacturing shows its first positive break since 1950 it comes decidedly *after* the break in the growth of aggregate GDP. Therefore we believe that focusing on the manufacturing sector as the engine of growth and the identification of ‘market reform’ (see Virmani 2006) as the stimulant is mis-guided.

6. We next turn to the estimates with regard to the Tertiary sector. The NAS disaggregates the Tertiary sector into four: ‘Trade, Hotels and Restaurants’, ‘Transport, Storage and Communications’, ‘Finance, Insurance and Real Estate’, and ‘Community and Personal Services’. Note from Table 2 that all these groups show an acceleration in the seventies, mostly prior to 1978-79 the date for the break in growth of GDP or, allowing for a confidence interval, co-terminus with this break itself. Of course, the extent of the acceleration varies from mild in the case of ‘Transport, Storage and Communication’ to dramatic in the case of ‘Financing, Insurance, Real Estate and Business Services’ (Table 2). We consider this evidence of a break in the components of the Tertiary sector as *prima facie* evidence of services having led the acceleration in the growth of GDP in India. Following this lead we undertake two exercises. First, we perform a decomposition of the contribution by sector to the change in the aggregate GDP growth rate from 1978-79. Secondly, we perform an econometric test of both the hypothesis of a services-led growth and of the associated relationships that we would expect to find if this were true.

[Table 3 goes about here]

Using the expression derived in the appendix, the change in the average growth rate of aggregate GDP from 1978-79 onwards is decomposed into percentage contributions by constituent sectors. The results of this exercise are reported in Table 3. Find that the contribution by the Tertiary sector overwhelms the contribution by the other sectors. In particular, note that the contribution exceeds that of manufacturing by a factor exceeding four. The above exercise appears conclusively to have established that the acceleration of aggregate GDP was not led by manufacturing. Methodologically, we have not reached this stage yet, for the results of the decomposition undertaken must be read along with the caveat that what may be established from the exercise is only the direct contribution of each sector to the change in the aggregate growth rate. *Per se* we cannot rule out that Tertiary-sector growth may have been in turn stimulated by manufacturing growth, even though our earlier finding is that services growth had broken ground before manufacturing. Taking into account this weakness of the decomposition exercise, we undertake some econometric exercises the results of which are presented in Table 4. For the purpose of these exercises we have aggregated the data according to the standard classification ‘Primary’, ‘Secondary’ and ‘Tertiary’ as this makes the analysis of the data more manageable. For a clear description of what each of these sectors contain the reader may refer to Table 3.

The regressions presented in Table 4 are the parsimonious representation of an original specification in which output growth in each sector was regressed on the current values of output growth in the other sectors and its own lagged values. Allowing for simultaneity the model was first estimated by GMM using instrumental variables for all current-dated variables. Except for the secondary sector in which the current growth of the tertiary sector was found to be statistically significant, for the primary and tertiary sectors the current output growth of the other sectors was found not to matter. So, for the sake of uniformity, we chose to exclude current output growth from the specification altogether. With current-dated variables out of the specification OLS is sufficient, and the resulting estimates are presented¹² in Table 4.

Before turning to the results a clarification regarding the final specification is in order. As the primary, secondary and tertiary sectors together exhaust GDP they have been excluded from the specification as causality is in question and simultaneity an estimation issue. Therefore values of these variables from lag one to three were entered along with the lagged

¹² All results reported here are available from the authors upon request.

dependent variable. The model was then reduced to the more parsimonious representation reported in the Table 4.

[Table 4 goes about here]

From the first regression note that the Tertiary sector alone of the three sectors is seen to drive GDP growth from an economic rather than an accounting point of view. It is to discern this economic role of sectoral growth that the independent variables have been entered in lags. For now we have evidence of ‘services as the principal engine of growth’. Next, note from the second regression that Secondary-sector output does not enter the equation for Tertiary-sector growth, while Primary-sector growth does. Finally, find from the third regression that services does enter into the determination of the growth of the Secondary sector. Together regressions two and three lead us to conclude that Secondary-sector growth has not driven services growth during the period under consideration, so it could not have influenced the growth shifts in the Indian economy even indirectly. We wager that it will be found that ‘the Tertiary-sector Granger-causes the Secondary-sector’ but ‘the Secondary sector does not Granger-cause the Tertiary’. Indeed, our results suggest a recursive model of growth.

IV. Conclusion

Using a method developed by Bai and Perron we have identified growth regimes in India since 1950. The hallmark of this method is that no extraneous information is imposed on the data, i.e., in the language of econometric practice we have ‘allowed the data to parametrise the model’. Our results suggest the following data-congruent narrative. There is across-the-board dynamism in the Indian economy during this period in that all the major sectors show an acceleration in their rate of growth. This radically diverges from the finding of the only other comparable study, by Wallack, that no break in the growth rate can be established for individual sectors. It implies that the acceleration in the economy-wide growth rate from 1978-79 identified in this study is not an artifact of the re-allocation of output across sectors. However, at least two of the three main components of GDP accelerate prior to that date giving us some idea of the factors underlying the acceleration in aggregate growth. These are the Primary and Tertiary sectors, or at least their major components. The Secondary sector, largely constituted by Manufacturing, shows a positive break only after 1980.

These results provide some vital clues and building blocks for a narrative on Indian economic growth. The sequence of accelerations, the order being primary, tertiary and secondary, suggests the following story of growth in India since 1950. The acceleration of

growth in the Primary sector, occurring as early as the mid-sixties provided the original stimulus, via supply and demand linkages, to growth in the other sectors of the economy. This is confirmed by our econometric exercise. Growth of the primary sector enters into the growth of the tertiary sector which in turn enters into the growth of the Secondary sector. The Secondary sector does not in turn enter either of the other; nor does the Tertiary sector enter into the growth of the Primary sector. We have a recursive model of growth in the economy here and a certain causality is implied by these results.

After a propulsion driven by acceleration of primary sector growth services growth has been unstoppable in India. As our decomposition exercise shows they contributed overwhelmingly to the estimated shift in the aggregate growth rate since 1978-79. Manufacturing has very likely played less than an active role in bringing about the growth transition in India as has been argued by some researchers. It accelerates after the acceleration in the other sectors which between them manage to raise the economy-wide growth rate well before the market reform said to have been initiated in the eighties. The results of our research strategy indicate little role for trade and industrial policy having been the trigger of a new growth dynamic in India via faster manufacturing growth, at least upto the mid-nineties. We conclude with two observations. Note that our results are a direct outcome of our methodology of generating the break date(s) endogenously. A very different reading would emerge were we to break the data arbitrarily ostensibly following a judgement¹³ on a shift in the policy regime which is believed to have stimulated growth. Clearly, therefore, methodology matters. We consider this an important result in itself. Secondly, some researchers have rhetorically queried whether services may emerge as the engine of economic growth in the future (Dasgupta and Singh, 2005.) We find that services have proved to be the engine of growth already! To recognise this is the prerequisite to an understanding of economic growth in India. However, the precise role, if any, of policy with respect to the accelerating growth of services remains to be analysed yet.

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¹³ See Acharya, 2004 and Kohli, 2006 for instances. Indeed the decade-wise periodisation leaves the researcher with little option other than to conclude that the acceleration of growth in India in the 1980s has been strongly led by faster growth of Agriculture (the main component of the Primary sector). For, agricultural growth not only doubles in the decade of the eighties but this acceleration is greater by far than that in the other two principal sectors of the economy (see Acharya, 2004, Table 2). In 1980-81 the Primary sector is the largest sector.

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Appendix

A1. Decomposition of change in aggregate growth rate

Consider two periods 1 & 2, respectively consisting of $n1$ and $n2$ years. Denote aggregate GDP, primary, secondary, and tertiary sector incomes by Y , P , S , and T . Let w_{it} denote the share of sector i ($i = P, S, T$) in Y in year t and g denote the growth rate. Y in year t can be written as follows:

$$Y_t = P_t + S_t + T_t \quad (1) \text{ From}$$

m (1), the growth rate of Y can be written as:

$$g_{Yt} = w_{Pt}g_{Pt} + w_{St}g_{St} + w_{Tt}g_{Tt} \quad (2) \text{ where}$$

ere $g_{Yt} = \left(\frac{Y_t - Y_{t-1}}{Y_{t-1}} \right)$ is the growth rate of Y in year t and g_{Pt} , g_{St} and g_{Tt} are defined

similarly. Using (2), the average annual growth rate for the period 1 ($\bar{g}_{Y1}$) can be written as follows. Where a bar over notations has been used to denote average value.

$$\bar{g}_{Y1} = \overline{w_{P1}g_{P1}} + \overline{w_{S1}g_{S1}} + \overline{w_{T1}g_{T1}} \quad (3)$$

where,

$$\bar{g}_{Y1} = n1^{-1} \left(\sum_{t=1}^{n1} g_{Yt} \right), \text{ and } \overline{w_{i1}g_{i1}} = n1^{-1} \left(\sum_{t=1}^{n1} w_{it}g_{it} \right).$$

Similarly, $\bar{g}_{Y2}$ can be written as below,

$$\bar{g}_{Y2} = \overline{w_{P2}g_{P2}} + \overline{w_{S2}g_{S2}} + \overline{w_{T2}g_{T2}} \quad (4)$$

In (4), averages are taken over $n2$ observations in period two. Subtracting (3) from (4),

$$(\bar{g}_{Y2} - \bar{g}_{Y1}) = (\overline{w_{P2}g_{P2}} - \overline{w_{P1}g_{P1}}) + (\overline{w_{S2}g_{S2}} - \overline{w_{S1}g_{S1}}) + (\overline{w_{T2}g_{T2}} - \overline{w_{T1}g_{T1}}) \quad (5)$$

Dividing (5) throughout by the left-hand side value and multiplying it by 100 gives the percentage contribution of each sector to the change in the average growth rate of Y in period two over that in previous period.

A2. Computation of the confidence interval of the breakdates.

Confidence interval for breakdates, where trended regressors are present, are computed using the procedure given in Bai (1997). This can be explained as follows. Let $\hat{T}_j$ be the j^{th} break

point, $\hat{\beta}_j = (\hat{a}_j, \hat{g}_j)$, $\hat{\delta}_j = (\hat{\beta}_{j+1} - \hat{\beta}_j)$ and $z_{tj} = (1, t_j)$, where t_j is the value of t at j^{th} break.

Define $\hat{L}_j = (\hat{\delta}_j' z_{tj}' z_{tj} \hat{\delta}_j) / \hat{\sigma}_u^2$, where $\hat{\sigma}^2 = T^{-1} \sum_{t=1}^T \hat{u}_t^2$ from (1). Now the 100(1- α)%

confidence interval is given by $[\hat{T}_j - (c / \hat{L}_j) - 1, \hat{T}_j + (c / \hat{L}_j) + 1]$, where c is the $(1-\alpha/2)^{th}$ quantile of the distribution (at 5 per cent level $c = 1.96$).

Table 1: Estimated breakdates

		First break	Second break	Third break	Fourth break
	GDP	1978-79 [1977/78-1979/80]			
1.1	Agriculture	1964-65 [1963/64-1965/66]			
1.2	Forestry and logging	1964-65 [1963/64-1965/66]	1976-77 [1975/76-1977/78]	1988-89 [1987/88-1989/90]	
1.3	Fishing	1958-59 [1957/58-1959/60]	1973-74 [1972/73-1974/75]	1982-83 [1981/82-1983/84]	1992-93 [1991/92-1993/94]
2	Mining and Quarrying	1959-60 [1958/59-1960/61]	1969-70 [1968/69-1970/71]	1980-81 [1978/79-1980/81]	1988-89 [1988/89-1990/91]
3	Manufacturing	1965-66 [1964/65-1966/67]	1982-83 [1981/82-1983/84]	1994-95 [1993/94-1995/95]	
3.1	Registered	1965-66 [1964/65-1966/67]	1982-83 [1981/82-1983/84]	1994-95 [1993/94-1995/96]	
3.2	Unregistered	1964-65 [1963/64-1965/66]	1986-87 [1985/86-1987/88]	1994-95 [1993/94-1995/96]	
4	Electricity, Gas and Water supply	1957-58 [1956/57-1958/59]	1967-68 [1966/67-1968/69]	1983-84 [1982/83-1984/85]	1991-92 [1990/91-1992/93]
5	Construction	1969-70 [1968/69-1970/71]	1981-82 [1980/81-1982/83]		
6	Trade, hotels and restaurants	1962-63 [1961/62-1963/64]	1971-72 [1970/71-1972/73]	1994-95 [1993/94-1994/95]	
6.1	Trade	1962-63 [1961/62-1963/64]	1971-72 [1970/71-1972/73]	1994-95 [1993/94-1995/96]	
6.2	Hotels and Restaurants	1959-60 [1951/52-1967/68]	1971-72 [1970/71-1972/73]	1984-85 [1983/84-1985/86]	1994-95 [1993/94-1995/96]
7	Transport, Storage and Communication	1961-62 [1960/61-1962/63]	1973-74 [1972/73-1974/75]	1993-94 [1992/93-1994/95]	

Note: 95 per cent confidence interval of the estimated break date is provided in the brackets. Confidence intervals are computed using the asymptotic distribution derived using shrinking magnitude of shift framework.

Table 1 Break-point Estimates ...Continues

		First break	Second break	Third break	Fourth brake
7.1	Railways	1961-62 [1960/61-1962/63]	1984-85 [1983/84-1985/86]	1992-93 [1991/92-1993/94]	
7.2	Transport by other means	1957-58 [1955/56-1959/60]	1973-74 [1972/73-1974/75]	1993-94 [1992/93-1994/95]	
7.3	Storage	1974-75 [1973/74-1975/76]	1984-85 [1983/84-1985/86]		
7.4	Communication	1962-63 [1960/61-1964/65]	1987-88 [1986/87-1988/89]	1995-96 [1994/95-1996/97]	
8	Financing, Insurance, real estate & business services	1966-67 [1965/66-1967/68]	1979-80 [1978/79-1980/81]	1992-93 [1991/92-1993/94]	
8.1	Banking & Insurance	1965-66 [1958/59-1972/73]	1979-80 [1978/79-1980/81]	1995-96 [1994/95-1996/97]	
8.2	Real estate, ownership of dwelling & business services	1966-67 [1965/66-1967/68]	1980-81 [1979/80-1981/82]	1993-94 [1992/93-1994/95]	
9	Community and Personal services	1961-62 [1960/61-1962/63]	1976-77 [1975/76-1977/78]	1992-93 [1991/92-1993/94]	
9.1	Public administration and defence	1961-62 [1960/61-1962/63]	1986-87 [1985/86-1987/88]	1995-96 [1994/95-1996/97]	
9.2	Other services	1957-58 [1956/57-1958/59]	1968-69 [1967/68-1969/70]	1980-81 [1979/80-1981/82]	1992-93 [1991/92-1993/94]

Note: 95 per cent confidence interval of the estimated break date is provided in the brackets. Confidence intervals are computed using the asymptotic distribution derived using shrinking magnitude of shift framework.

Table 2. Growth rate Estimates

	Sector	Period 1	Period 2	Period 3	Period 4	Period 5
	GDP	3.34 [1950/51-1977/78]	5.20 [1978/79-2003/04]			
1.1	Agriculture	1.73 [1950/51-1963/64]	2.78 [1964/65-2003/04]			
1.2	Forestry and logging	1.57 [1950/51-1963/64]	2.46 [1964/65-1975/76]	-2.31 [1976/77-1987/88]	1.33 [1988/89-2003/04]	
1.3	Fishing	5.63 [1950/51-1957/58]	3.93 [1958/59-1972/73]	2.14 [1973/74-1981/82]	6.18 [1982/83-1991/92]	5.10 [1992/93-2003/04]
2	Mining and Quarrying	4.82 [1950/51-1958/59]	5.44 [1959/60-1968/69]	3.14 [1969/70-1979/80]	8.65 [1980/81-1987/88]	4.90 [1988/89-2003/04]
3	Manufacturing	6.27 [1950/51-1964/65]	4.26 [1965/66-1981/82]	6.76 [1982/83-1993/94]	6.07 [1994/95-2003/04]	
3.1	Registered	7.76 [1950/51-1964/65]	4.53 [1965/66-1981/82]	7.53 [1982/83-1993/94]	6.13 [1994/95-2003/04]	
3.2	Unregistered	4.63 [1950/51-1963/64]	4.09 [1964/65-1985/86]	5.96 [1986/87-1993/94]	5.83 [1994/95-2003/04]	
4	Electricity, Gas and Water supply	8.51 [1950/51-1956/57]	12.21 [1957/58-1966/67]	7.03 [1967/68-1982/83]	9.23 [1983/84-1990/91]	5.64 [1991/92-2003/04]
5	Construction	6.59 [1950/51-1968/69]	2.00 [1969/70-1980/81]	5.05 [1981/82-2003/04]		
6	Trade, hotels and restaurants	5.79 [1950/51-1961/62]	3.80 [1962/63-1970/71]	5.12 [1971/72-1993/94]	8.40 [1994/95-2003/04]	
6.1	Trade	5.71 [1950/51-1961/62]	3.84 [1962/63-1970/71]	5.10 [1971/72-1993/94]	8.23 [1994/95-2003/04]	
6.2	Hotels and Restaurants	5.42 [1950/51-1958/59]	4.67 [1959/60-1970/71]	4.45 [1971/72-1983/84]	6.74 [1984/85-1993/94]	9.91 [1994/95-2003/04]
7	Transport, Storage and Communication	6.12 [1950/51-1960/61]	5.50 [1961/62-1972/73]	5.55 [1973/74-1992/93]	9.21 [1993/94-2003/04]	

Notes: (i) All estimates are significant at one per cent level, and (ii) Period of the reported growth rate is provided in bracket.

Table 2. Growth rate Estimates...Continues

		Period 1	Period 2	Period 3	Period 4	Period 5
7.1	Railways	5.81 [1950/51-1960/61]	3.68 [1961/62-1983/84]	4.01 [1984/85-1991/92]	3.78 [1992/93-2003/04]	
7.2	Transport by other means	5.94 [1950/51-1956/57]	6.16 [1957/58-1972/73]	6.21 [1973/74-1992/93]	6.85 [1993/94-2003/04]	
7.3	Storage	3.09 [1950/51-1973/74]	9.99 [1974/75-1983/84]	1.15 [1984/85-2003/04]		
7.4	Communication	7.13 [1950/51-1961/62]	6.17 [1962/63-1986/87]	7.41 [1987/88-1994/95]	19.41 [1995/96-2003/04]	
8	Financing, Insurance, real estate & business services	3.06 [1950/51-1965/66]	3.44 [1966/67-1978/79]	8.88 [1979/80-1991/92]	8.00 [1992/93-2003/04]	
8.1	Banking & Insurance	6.65 [1950/51-1964/65]	5.83 [1965/66-1978/79]	11.01 [1979/80-1994/95]	9.19 [1995/96-2003/04]	
8.2	Real estate, ownership of dwelling & business services	2.29 [1950/51-1965/66]	2.79 [1966/67-1979/80]	7.77 [1980/81-1992/93]	5.92 [1993/94-2003/04]	
9	Community and Personal services	4.20 [1950/51-1960/61]	4.46 [1961/62-1975/76]	5.95 [1976/77-1991/92]	6.86 [1992/93-2003/04]	
9.1	Public administration and defence	6.53 [1950/51-1960/61]	6.23 [1961/62-1985/86]	4.96 [1986/87-1994/95]	6.48 [1995/96-2003/04]	
9.2	Other services	2.41 [1950/51-1956/57]	3.97 [1957/58-1967/68]	2.83 [1968/69-1979/80]	5.01 [1980/81-1991/92]	7.43 [1992/93-2003/04]

Notes: (i) All estimates are significant at one per cent level, and (ii) Period of the reported growth rate is provided in bracket.

Table 3. Sectoral contribution to the change in GDP growth from 1978-79

		Contribution in %
I	Primary Sector	-25.95
1.1	Agriculture	-26.41
1.2	Forestry and logging	-3.94
1.3	Fishing	0.63
2	Mining and Quarrying	3.76
II	Secondary Sector	25.66
3	Manufacturing	24.65
3.1	Registered	18.79
3.2	Unregistered	5.86
4	Electricity, Gas and Water supply	4.44
5	Construction	-3.43
III	Tertiary Sector	100.29
6	Trade, hotels and restaurants	23.08
6.1	Trade	20.83
6.2	Hotels and Restaurants	2.25
7	Transport, Storage and Communication	19.83
7.1	Railways	-0.67
7.2	Transport by other means	7.58
7.3	Storage	-0.12
7.4	Communication	13.04
8	Financing, Insurance, real estate & business services	37.24
8.1	Banking & Insurance	21.74
8.2	Real estate, ownership of dwelling & business services	15.50
9	Community and Personal services	20.14
9.1	Public administration and defence	7.86
9.2	Other services	12.28

Note: The table provides the percentage contribution of sectors to the increased growth of GDP from 1978-79.

Table 4. Regression Results

Explanatory variables	Dependent variable			
	GDP _t	P _t	S _t	T _t
P _{t-1}	0.055 (1.30)	-0.596* (-5.25)	0.060 (1.36)	0.066* (2.23)
S _{t-1}	0.139 (1.09)	-0.224 (-0.77)	0.168 (1.19)	-0.068 (-1.12)
T _{t-1}	1.112* (3.74)	0.307 (0.47)	0.429* (2.16)	0.524* (3.58)
GDP _{t-1}	-0.769* (-3.45)			
GDP _{t-2}	-0.406* (-2.79)			
P _{t-2}		-0.219 (-1.72)		
S _{t-2}			-0.198 (-1.56)	
T _{t-2}				0.265* (2.19)
$\overline{R}^2$	0.16	0.24	0.10	0.49
Breusch-Godfrey LM test for Autocorrelation				
First order	0.56 (0.45)	2.02 (0.16)	0.28 (0.59)	0.07 (0.79)
Second order	2.07 (0.36)	3.17 (0.21)	0.36 (0.83)	1.88 (0.39)
No. Observations	50	50	50	50

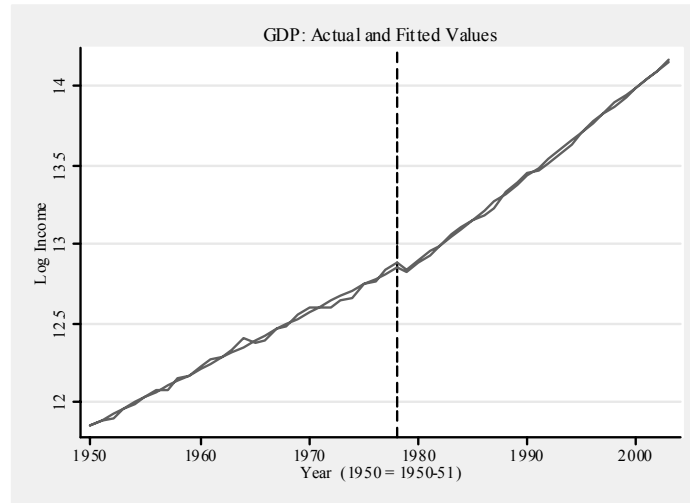
Notes:

(1) Where GDP, P, S, and T, respectively, denote the growth rate of GDP and Primary, Secondary and Tertiary sector output.

(2) *t* values, computed using Huber/White/sandwich estimator of variance, are given in parentheses.

(3) For Breush-Godfrey LM test Chi-Square values are reported and parentheses provide the P value.

(4) * indicates significant at five per cent level.

Figure A1.

Note: The vertical dashed line indicates the break year.